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Massy, Île-de-France, France

Mohamed Amine AROUS



AI Engineer | Generative AI, LLMs, RAG & AI Agents

AI Engineer specializing in Generative AI, with experience developing and deploying solutions based on Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), AI Agents, and Knowledge Graphs. Experience at CEA List, École Polytechnique, and STMicroelectronics. Skilled in Python, FastAPI, AWS, Docker, and AI system evaluation, with a particular focus on reliability and production deployment.

PROFESSIONAL EXPERIENCE

CEA List (LASTI Team) – Research Intern | AI Engineering & Knowledge Graphs

Apr 2026 – Oct 2026

Supervisor: Prof. Gaël De Chalendar

Palaiseau, France

- Developed a *Neuro-Symbolic AI* pipeline combining *Foundation Models* and an *enriched Knowledge Graph* for semantic understanding of scenes involving autonomous warehouse robots.
- Designed and implemented a *Knowledge Graph enrichment pipeline* with 89% accuracy; evaluated the reliability of the injected semantic knowledge.

CMAF Laboratory, École Polytechnique – Research Intern

May – Sep 2025

Supervisor: Prof. Eric Moulines

Palaiseau, France

- Developed *CosyVoice2-EU*, an open-source adaptation of a *Transformer-based voice cloning* model for European languages ([GitHub](#)).
- Achieved performance *exceeding the best open-source baselines* while remaining *competitive with a commercial speech synthesis baseline (ElevenLabs)*, contributing to a *paper accepted at EUSIPCO*.

PROJECTS

Portfolio with RAG-Powered AI Assistant

2026

[Frontend GitHub](#) [Backend GitHub](#)

[Live](#)

- Developed a *serverless* assistant widget for my portfolio using *FastAPI* on *AWS Lambda* and hybrid *BM25/embeddings* search to answer visitors' questions based on my career data.
- Designed for near-zero operating costs and validated with *413 passing tests*, with zero hallucinations across 18 evaluation questions.

NORMA – Knowledge Graph & GraphRAG for Automotive Regulations [GitHub](#)

2026

Normative Ontology for Requirements Modeling & Analysis

[Live](#)

- Developed a bilingual *knowledge graph* pipeline transforming *UNECE/EU* regulations into a versioned, traceable *Neo4j* graph using an *OWL* ontology; *SHACL* validation reduced 21,700 violations to 0.
- Deterministic link expansion improved multi-hop *Recall@10* by $2.2\times$ ($0.317 \rightarrow 0.708$). Deployed on *AWS Lambda*, ~\$0.003/month.

AWS CI/CD Deployment Pipeline [GitHub](#)

2026

- Designed and deployed a fully automated *CI/CD pipeline* on *AWS*, taking a web application from Git commit to production without any manual steps.
- Automated artifact management, builds, deployment, and release orchestration using *CodePipeline*, *CodeBuild*, *CodeDeploy*, and *CodeArtifact*.
- Deployed to production on *EC2*: releases secured by *IAM*, triggered on every GitHub push.

Embodied AI with Vision-Language-Action Models (VLAs) [GitHub](#)

Oct 2025 – Mar 2026

Academic project, team of 5, supervised by Prof. Gianni FRANCHI

- Developed an *automated benchmarking framework* to evaluate the AIUTA agent on *multiple LLM backbones (Qwen3-8B, Llama3-8B)* on a *GPU cluster*.

Veo Hackathon | Team of 6

- Developed *AI agents* with *CrewAI* and *MistralAI* to automate candidate screening through *SWOT analysis* of CVs and evaluation of *body language* in video interviews.
- Automated candidate communication and report delivery using *Python* and *SMTP* email services.

Team Engineering Project

Oct 2024 – May 2025

STMicroelectronics

Team project

- Developed *machine learning models* in *Python* (*Scikit-learn*, *PyTorch*, *Pandas*, *NumPy*) to evaluate similarity between microelectronic components.
- Achieved 98% accuracy, *outperforming the company's internal baseline solution*.
- Collaborated within a multidisciplinary engineering team to design and validate the final solution.

EDUCATION

Engineering Degree, Artificial Intelligence Specialization (Double Degree)

ENSTA Paris – ENIT

2023 – 2026

2nd year: GPA 3.96/4.3 | 1st year: Ranked 2/50 (ENIT)

Coursework: Deep Learning, Machine Learning, Language Modeling, Reinforcement Learning, game theory & multi-agent control, probabilistic reasoning

Preparatory Classes in Mathematics and Physics

Institut Préparatoire aux Études d'Ingénieurs de Tunis (IPEIT)

2021 – 2023

Ranked TOP 4/177 (2nd year) | TOP 3/326 (1st year)

Coursework: linear algebra, probability, statistics, numerical analysis, physics, chemistry, Python programming, databases (SQL)

TECHNICAL SKILLS

- **Programming Languages:** Python, C++, SQL
- **Generative AI:** Large Language Models (LLMs), Retrieval-Augmented Generation (RAG), AI Agents, Agentic AI, Transformers, Vector Embeddings, Semantic Search, Ollama, CrewAI
- **Machine Learning:** PyTorch, TensorFlow, Scikit-learn, CUDA, Model Evaluation
- **Backend & APIs:** FastAPI, REST APIs, Flask, Streamlit, Gradio
- **Data & Semantic AI:** PostgreSQL, Neo4j, DynamoDB, Knowledge Graphs, BM25, RDF, OWL, SHACL, Named Entity Recognition (NER)
- **Cloud & DevOps:** AWS, AWS Lambda, EC2, IAM, CodePipeline, CodeBuild, CodeDeploy, CodeArtifact, Docker, CI/CD, Git, GitHub

PUBLICATIONS

- “Europeanizing Modular Zero-Shot TTS: Component-Level Adaptation Framework for French & German.” EUSIPCO 2026. [PDF](#) [Code](#)

AWARDS & DISTINCTIONS

Excellence Scholarship (Tunisian Government) · **Google MENA ML 2026** (full scholarship, Google DeepMind) · **School Ambassador (5x)** (competitive programming & AI)

CERTIFICATIONS (ALL CERTIFICATIONS)

- **AWS Certified AI Practitioner** – Amazon Web Services, 2026 [Verification](#)
- **Generative AI Engineering Professional Certificate** – IBM, 2025 (16-course program) [Verification](#)
- **Data Science Professional Certificate** – IBM, 2025 (6-course program) [Verification](#)
- **Foundations of AI Professional Certificate** – IBM, 2025 (3-course program) [Verification](#)

VOLUNTEERING & LEADERSHIP

Tutor – #GENIUS Program, École Polytechnique — weekly academic support in science & mathematics · **Competitive Programming Lead**, G²FOSS — organized DSA courses and competitions

LANGUAGES

English (Linguaskill: **C1**) | French (TCF: **C1**) | Arabic (**Native**)